Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Eaton, John Ph.D.		
Project Title:	Attenuation of Gas Turbulence by a Nearly Stationary Dispersion of Fine Particles		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305-3030	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	01/12/2002	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Eaton, John		
COI Name (Institution):			
Grant/Contract No.:	NCC3-640		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	Dispersed particles have been known to modify the gas phase statistics in turbulent gas flows. Under certain conditions the turbulence is attenuated. It is still not clear what mechanisms are responsible though. One of the parameters that makes the problem complex is gravity. The particles fall through the turbulent eddies due to gravity and create mean wakes, thereby distorting the flow structure. By eliminating this parameter the problem can be greatly simplified, and the direct interactions between fluid and particle can be examined. A turbulence chamber that can create homogeneous and isotropic turbulence with no mean flow has been developed. It will be used in terrestrial and microgravity environments. The particles enter and fall through the chamber under terrestrial conditions and float in the chamber under microgravity conditions. Comparing the results from these two test cases will hopefully shed some light on the underlying physics of turbulence attenuation by particles and also create a database for model developers. Turbulence attenuation measurements will be made in two environments. The first is under terrestrial conditions where the particles fall through the turbulent eddies due to gravity. The second is in a microgravity environment where the mean particle wakes are eliminated and the particles are stationary. 1. Gravity causes particles to fall through the turbulent eddies in turbulent particle-laden flows. A microgravity environment will make the particles stationary, giving us a better understanding of this complex phenomenon. A turbulence chamber capable of creating turbulence with high Taylor microscale Reynolds numbers and no inflow or outflow is ideal for use in a microgravity environment where space is limited. The chamber that has been developed can create homogeneous and isotropic turbulence, and it will be able to provide a database for ongoing numerical simulations of particle-laden turbulent flows. Such a chamber could also be used for other turbulence research being conducted in a microgravity environment. 2. Measurements in a particle-laden flow must be nonintrusive. A particle image velocimetry (PIV) system has been developed that can provide information on the structure of the turbulence in such flows. It can measure statistics such as two-point velocity correlations, Reynolds stresses, and instantaneous velocity gradients. 3. The measurements will be the main product of the research. They will be used to test various models for turbulence modification in particle-laden flows. We believe it will be able to play a significant role in resolving controversy about the mechanisms and modeling of turbulence attenuation by particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 08/22/2007)